

The Health of the Commons: Relational Health and Intelligence Across Persons, AI Systems, and Shared Conditions

Verdict — Synthesis. A defensible common interface is possible, but the strongest version of the hypothesis should be weakened. I did **not** find a validated instrument that measures people, AI systems, institutions, relations, and their commons on one shared scale. I did find unusually close pieces. The WHO ICF separates functioning from environmental facilitators and barriers and distinguishes capacity in a standardized environment from performance in the actual one; the capability approach treats resources as being converted into real opportunities through personal, social, and environmental factors; relational-wellbeing work explicitly treats wellbeing as something produced between persons, collectives, institutions, and material conditions; health economics measures family spillovers; OECD and inclusive-wealth frameworks separate current wellbeing from stocks that sustain future wellbeing; Ostrom studies the governance of shared resources; Piovarchy and Siskind explicitly apply **epistemic health** to persons, communities, and nations; network and risk-sharing studies show that relations can transmit information, support, risk, and harm; and AI research separately measures calibration, correction, data provenance, recursive-data degradation, human labour, and resource use. ¹ The proposed contribution is therefore **not a new discovery that functioning is relational**. What appears not to exist in the literature searched is a common measurement *grammar* that links a node's capacity, its incoming support and information, the state of shared conditions, its outward effects, and the reach of those effects across humans, AI systems, and institutions. The most important correction to the brief is that “stock” is too narrow: climate state, institutional capacity, a care workforce, a body of verified knowledge, political stability, and an information network do not behave as one fungible stock. I recommend **shared conditions/state** as the umbrella term. I also recommend five functional domains—**grounding, adaptive learning with retention, correction with independent routes, resource viability, and resilience/recovery**—while treating **compression/efficiency** as a cross-cutting cost property and **contribution/outflow/reach** as relational accounting, not intrinsic health. A profile, thresholds, and explicit interactions are preferable to a sum or product. Most importantly, a person's claim to support must never depend on their measured contribution.

Label key. **Established** means a framework or empirical result confirmed against a retrieved publisher, DOI, PubMed, official institutional, or repository record. **Synthesis** means my interpretation or design recommendation. **Hypothesis** means an empirical claim that remains to be tested. The EbE repository is treated throughout as the theory under assessment, not as external evidence.

Prior art and what appears to be missing

Established. The closest established framework on the human side is probably the **International Classification of Functioning, Disability and Health (ICF)**. The ICF says functioning and disability occur in context and explicitly includes environmental factors. Its activities-and-participation coding distinguishes *performance* in the person's current environment from *capacity*, the highest probable functioning in a standardized environment. Environmental factors can be coded as facilitators or barriers. This is strikingly

close to the brief's distinction between own capacity and supported functioning. It is not, however, a theory of commons health, reciprocal outflow, or AI functioning. ²

Established. The **capability approach** is the closest conceptual ancestor of the proposed transformation equation. It separates resources from what a person can actually be and do. Robeyns' account distinguishes capabilities from achieved functionings and emphasizes that conversion from means to functionings varies between people and circumstances; later capability literature explicitly organizes conversion factors as personal, social, and environmental. This makes "own capacity $\times$ inflow $\times$ shared conditions" less novel than it initially looks. Yet capability theory is primarily a normative framework about substantive freedoms and valuable functionings, not an engineering account of reciprocal flows or a common health measure across humans and machines. ³

Established. Sarah White's **relational wellbeing** framework comes still closer to the ontology. It treats wellbeing as a social process with material, relational, and subjective dimensions, measurable at individual and collective levels, and explicitly describes wellbeing as occurring in relations between individual and collective, local and global, and people and state. Subsequent relational-wellbeing work treats persons as relational subjects, relationships as conduits of resources and power, and context as part of how wellbeing is generated. That is strong prior art against any claim that relationality itself is new. ⁴

Established. Health economics already recognizes that an intervention directed at one patient can alter the wellbeing of family members and carers. Al-Janabi and colleagues developed a framework for family health spillovers; subsequent economic evaluations have shown that including family effects can alter cost-effectiveness results. The historical review of the QALY supplied with this brief likewise identifies family and carer spillovers, wider-than-health outcomes, capability measures, and equity concerns as important unresolved issues in the QALY tradition. ⁵ Carer-burden instruments such as the Zarit Burden Interview make the other side of support visible: care is not a costless external input. Validated versions correlate caregiver burden with poorer psychological outcomes. ⁶

Established. At the societal level, OECD wellbeing accounting and inclusive-wealth approaches already distinguish **current outcomes** from the **resources that sustain future outcomes**. OECD's wellbeing framework tracks current wellbeing alongside natural, human, economic, and social capital; inclusive-wealth accounting treats produced, human, and natural capital as a country's productive base for sustaining wellbeing. The Dasgupta Review similarly treats nature as an asset and focuses attention on what humans take from, transform within, and return to natural systems. These are clear precedents for distinguishing flow from sustaining conditions. ⁷

Established. Ostrom provides the strongest prior art for **shared-condition governance** rather than individual wellbeing. *Governing the Commons* shows that common-pool resources can sometimes be governed sustainably through locally evolved rules, monitoring, sanctions, conflict-resolution arrangements, participation, and nested governance rather than only privatization or centralized control. Hess and Ostrom explicitly extend commons analysis to knowledge, while noting an important difference: information is often non-rival, so its vulnerabilities concern production, verification, preservation, access, enclosure, and governance rather than simple physical depletion. ⁸

Established. The single closest use of the actual word *health* is Piovarchy and Siskind's **epistemic health**. They define the epistemic health of entities including persons, communities, and nations in terms of how well they function with respect to epistemic goods or ideals, such as true belief and reliable inference, and

discuss social trust and research institutions as influences on it. This is an important partial anticipation of “one concept read at several levels.” It is not a validated cross-level measurement instrument, does not cover resource viability or care, and does not establish equivalence across humans and AI systems. ⁹

Synthesis. The strongest novelty claim that survives is therefore narrower:

Provisional definition of the health of the commons: *the capacity of a bounded set of shared ecological, informational, institutional, social, and technical conditions to continue supporting the functioning of its dependent participants over a stated horizon, while preserving reliable contact with reality, learning from error, keeping routes for correction open, maintaining the resources on which that functioning depends, and recovering from disturbances without systematically exhausting particular contributors or excluded groups.*

This is a **functional definition**, not a claim that every commons is an organism. It also does not imply that an AI system has subjective wellbeing or moral status.

Synthesis. I found no verified prior instrument that simultaneously does all of the following: uses a common functional schema for humans, AI systems, and institutions; measures a node, its incoming relations, and its shared conditions; records outward effects and their reach; distinguishes support from intrinsic impairment without withdrawing necessary assistance; and links all layers empirically to future “staying in step.” Absence from this search is not evidence of global novelty.

Map of the literatures and the proposed dimensions

Synthesis. The table below maps what each literature primarily covers. It is deliberately about *coverage*, not equivalence. ● means central; ◐ means partial or adjacent; ○ means little direct coverage. The columns are: **L** learning/change, **E** compression/efficiency, **G** grounding/coherence, **C** correction/corrigibility, **V** contribution/resource viability, **I** inflow, **S** shared state or stock, **O** outflow, and **R** reach.

Literature	Main object / level	L	E	G	C	V	I	S	O	R
Capability approach	Person within social/ environmental conversion conditions	◐	○	○	○	◐	●	●	◐	◐
WHO ICF	Person/functioning + environment	○	○	○	○	◐	●	●	○	○
Social determinants of health	Population, institutions, living conditions	○	○	○	◐	●	●	●	○	◐
Carer/family spillovers	Dyad/family/network	○	○	○	○	●	●	◐	●	◐
Care ethics / relational autonomy	Person-in-relation; dependence and power	○	○	◐	●	●	●	◐	●	●
Social capital / collective efficacy	Relation, neighbourhood, community	◐	◐	◐	●	◐	●	●	●	●

Literature	Main object / level	L	E	G	C	V	I	S	O	R
Relational wellbeing	Person–relation–collective	◐	○	◐	◐	◐	●	●	●	●
Wellbeing / inclusive-wealth accounting	Society; present outcomes and future resources	◐	●	○	◐	●	◐	●	●	◐
Planetary health / doughnut-type frameworks	Human society within ecological limits	◐	●	◐	◐	●	◐	●	●	●
Ostrom / knowledge commons	Commons users, rules, resources	●	●	◐	●	●	●	●	●	●
Epistemic health / democracy / information ecosystem	Person, community, polity	●	◐	●	●	◐	●	●	●	●
Network contagion / risk-sharing	Nodes, dyads, networks	◐	●	◐	◐	●	●	◐	●	●
AI model health / AI commons	Model, pipeline, data and infrastructure ecosystem	●	●	●	●	●	●	●	●	◐

Established. Several findings behind this table matter directly. Social determinants research treats health as shaped by the conditions in which people are born, grow, live, work, and age and by wider political and economic forces. That strongly supports measuring context rather than attributing all realized functioning to the node. ¹⁰ Social capital and collective efficacy research likewise shows that relations have properties at a level above individuals. Sampson, Raudenbush, and Earls operationalized neighbourhood collective efficacy as social cohesion combined with willingness to intervene for the common good and found it associated with lower violence. At the same time, reviews document a “dark side” of social capital, including exclusion, contagion, and cross-level harms. More trust or tighter connection is therefore not automatically healthier. ¹¹

Established. Network epidemiology illustrates both the attraction and the danger of interpreting relational effects. Christakis and Fowler reported clustering and longitudinal associations in obesity through social ties in the Framingham network. The study itself discussed homophily and common exposure, and published critics immediately raised residual socioeconomic and network-homogeneity explanations. Network association is therefore not enough to identify an “outflow” from one node causing a downstream health state. ¹² Stronger causal evidence comes from randomized or quasi-random network studies, but those results are domain-specific. Economic experiments, for example, show that network position can support contract enforcement and risk sharing. ¹³

Established. Empirical evidence also supports the weaker proposition that shared conditions can affect cognitive and psychological functioning. Reviews find that extreme environmental stress, including heat, can impair some cognitive tasks, although effects vary by exposure, task, and study design. Higher allostatic load is associated with poorer global cognition and executive function, but the reported meta-analytic correlations are small and longitudinal evidence is less complete. Studies of loneliness and social isolation are heterogeneous: several longitudinal studies and a meta-analysis report increased cognitive risk, while

other cohorts find weak or null longitudinal associations after adjustment. This argues for modeling environment × person interactions, not assigning a universal “commons multiplier.” ¹⁴

Established. The informational environment offers clearer experimental examples. Randomized studies and meta-analyses show that exposure to health misinformation increases false belief on average, although effect sizes and moderators vary; repetition itself can increase perceived truth. Fact-checks can reduce targeted false beliefs immediately, but effects may decay. This makes information quality and correction routes plausible commons-level variables. ¹⁵

Established. Existing societal data can operationalize parts of the proposed profile. V-Dem version 16, released in March 2026, contains hundreds of indicators and indices spanning electoral, liberal, participatory, deliberative, and egalitarian democracy. OECD's Trust Survey measures perceived voice, reliability, responsiveness, fairness, integrity, evidence use, and trust in public institutions. These are useful indicators of correction and reach, although trust itself must not be treated as evidence that an institution is truthful. ¹⁶

Synthesis. The table exposes one weakness in the original five dimensions: **compression/efficiency is the least consistently supported as a dimension of health**. Efficiency clearly matters for persistence and affordability, but redundancy, spare capacity, diverse information routes, and duplicated checks are often valuable precisely because they are inefficient in the narrow sense. Adaptive-governance research stresses diversity, bridging organizations, learning, and redundancy during change. Treating maximal compression as health could therefore reward fragility. ¹⁷ I would retain efficiency as “resources consumed per retained capability” and test it against health outcomes, not place it on equal footing with grounding or resilience.

Testing inflow, shared state, outflow, reach, and one profile across levels

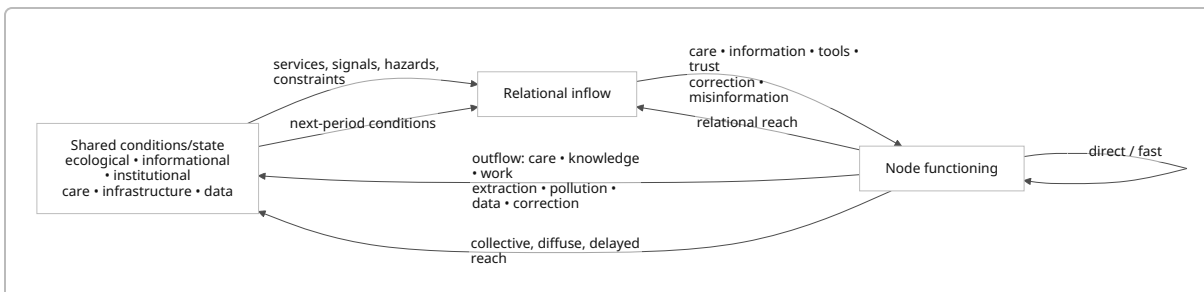
Established. The proposed **inflow** concept has strong antecedents. In health it appears as environmental facilitation, social support, service access, family care, socioeconomic conditions, and risk-sharing transfers. In the ICF, the same impairment may lead to very different performance depending on environmental facilitators and barriers. In the capability approach, the same resources yield different capabilities because conversion conditions differ. Informal-insurance research shows that network connections can smooth shocks, but insurance is incomplete and depends on network structure, commitment, enforcement, and capacity. ¹⁸

Synthesis. Shared state is preferable to “stock.” Some common conditions really are stocks: groundwater, infrastructure, financial reserves, verified datasets, or workforce capacity. Others are state variables or processes: political stability, norms, error-correction institutions, media diversity, or climate risk. Calling all of these “stock” invites a false conservation law and makes unlike things appear substitutable. A healthy wetland cannot in general compensate for a captured appeals court; plentiful compute cannot compensate for corrupted training provenance; high social trust cannot compensate for systematic misinformation.

Established. Outflow also has established analogues, but it is heterogeneous. Informal transfers can insure other households; caregiving improves another person's functioning while imposing time and health costs on the caregiver; participation can help maintain collective institutions; extraction can deplete rival resources; knowledge can be copied without depleting the original information while still imposing costs for

verification, curation, and maintenance. Ostrom's distinction between common-pool resources and knowledge commons is therefore essential. 19

Synthesis. The relational loop should therefore be written as a vector system, not one scalar ledger:



Synthesis. This is compatible with EbE's intuition but changes its interpretation. There need not be one common currency. Each arrow may be beneficial on one dimension and harmful on another. A caregiver can supply excellent care while losing sleep and income. An AI answer can be informationally useful while consuming energy. A policy can improve average service efficiency while closing an appeal route. The accounting needs **vectors, distributions, and time**, not just net flow.

Established. Something close to the proposed **reach** construct already exists, but no existing concept is quite the same. Self-efficacy concerns belief in one's capacity to perform actions; political efficacy distinguishes confidence in one's own political competence from belief that institutions will respond; collective efficacy concerns a group's shared capacity to act. OECD surveys likewise ask whether citizens believe they have a say. These are primarily perceptions or capabilities, not causal measures of how far an intervention actually propagates. 20

Synthesis. I would define **reach** as *the empirically attributable change a node can produce in specified targets, by level and time horizon*. Report at least three components: direct reach over the node's own state; relational reach over identifiable others; and commons reach over shared conditions. Also report **perceived reach** separately. Confusing the two is dangerous: a powerless person can feel responsible for a climate outcome they cannot individually alter, while a powerful institution can understate its actual externalities.

Synthesis. "One profile at three levels" is defensible only as a **common functional grammar**, not as one psychometric scale. Consider grounding. For a person it may mean whether confidence tracks accuracy on relevant judgments. For an AI system it may mean factual accuracy and calibrated uncertainty under distribution shift. For a polity it may mean whether administrative statistics, journalism, science, audit, and deliberation produce and preserve reliable evidence. These are related questions—*does the system remain coupled to what is actually happening?*—but not interchangeable observations.

Synthesis. I therefore recommend five shared functional questions:

Domain	Substrate-neutral question	Why retain it
Grounding	Does information about relevant external states reliably alter the system's representation and confidence?	Covers truth contact and calibration.
Adaptive learning and retention	Can valid new information produce useful change without erasing still-valid learning?	Separates plasticity from mere instability.
Correction and route independence	Can errors be challenged, transmitted, adjudicated, and corrected without requiring obedience to a single source?	Captures corrigibility plus independence.
Resource viability	Can required upkeep continue over the stated horizon without exporting unsustainable costs to dependencies?	Makes hidden depletion visible.
Resilience and recovery	After disturbance, can adequate functioning be restored within an acceptable time and cost?	Covers robustness without demanding rigidity.

Synthesis. Contribution, inflow, reach, and efficiency should accompany these five rather than be folded into them. That preserves the ethically crucial distinction between **being healthy enough to function** and **being useful to somebody else**.

Synthesis. On combination, a **profile comes first**. Multiplying ordinal scores such as 0–4 is not meaningful merely because the theory says bottlenecks matter; the numerical coding of ordinal categories is arbitrary. A product also lets measurement noise near zero dominate the result. A better research programme compares additive, interaction, minimum/threshold, and dynamic survival models on out-of-sample prediction of a preregistered outcome such as maintaining task accuracy, service continuity, or independent living over a stated horizon. A non-compensable **red-flag threshold** is reasonable where theory and data establish one—for example, no functioning correction channel—but that threshold should be empirically justified rather than assumed.

Formal link to transfers, viability, risk sharing, and caregiver depletion

Established about the repository, not external evidence. The two cited Lean results are present in `NetworkVortexLedger.lean`. The file defines a pure transfer as an internal flow that sums to zero across the declared set of parts. `internallyViable_iff_exists_viable_transfer` proves, under additive part aggregation and for a nonempty finite set of parts at a fixed indexed time, that nonnegative aggregate internal slack is equivalent to the existence of a pure transfer that leaves every part with nonnegative part slack. `transfer_breaks_part_with_whole_unchanged` supplies a two-part example in which moving one unit from a “commons” part to a taker makes the commons part nonviable while total slack remains 2. The source file itself states its limits: it is one-time, linear, additive accounting with a declared decomposition; it does not decide which real flows count as transfers, model downstream changes after a part fails, or derive normative conclusions. The corresponding prediction is also stated in `PREDICTIONS.md` as a claim to be tested, not empirical evidence.

Synthesis. Mathematically, the first result is closer to a **budget-balanced transfer feasibility statement** than to a novel theory of care or commons. With unrestricted real-valued transfers and additive slack, a nonnegative aggregate can be redistributed to make each component nonnegative. Real risk-sharing models add the difficult parts: incomplete information, network restrictions, transaction costs, limited commitment, strategic defection, social collateral, and unequal bargaining power. Ambrus, Mobius, and Szeidl, for example, show that network structure constrains informal insurance, while Townsend documents incomplete consumption insurance in real low-income settings. ²¹

Established. Viability theory is the deeper formal analogue for the dynamic question. Aubin and colleagues define viability in terms of whether trajectories can remain within admissible constraints under available controls and explicitly apply the framework across biological, economic, environmental, and control systems. Unlike the static EbE ledger theorem, viability kernels ask which starting states allow continued satisfaction of constraints over time. ²²

Synthesis. “The carer who burns out” therefore needs a dynamic theorem. A minimal version could use a caregiver reserve vector B_t :

$$B_{t+1} = B_t + U_t + S_t - M_t - C_t$$

where U_t is the caregiver’s own regenerative input, S_t is external support or respite, M_t is ordinary maintenance, and C_t is the cost of care supplied to another node. Care remains jointly viable only while both the recipient’s required support constraint and the caregiver’s reserve constraints are satisfied.

Hypothesis. If there is some reserve dimension k such that, over a sustained interval,

$$E[U_t^k + S_t^k - M_t^k - C_t^k] < 0,$$

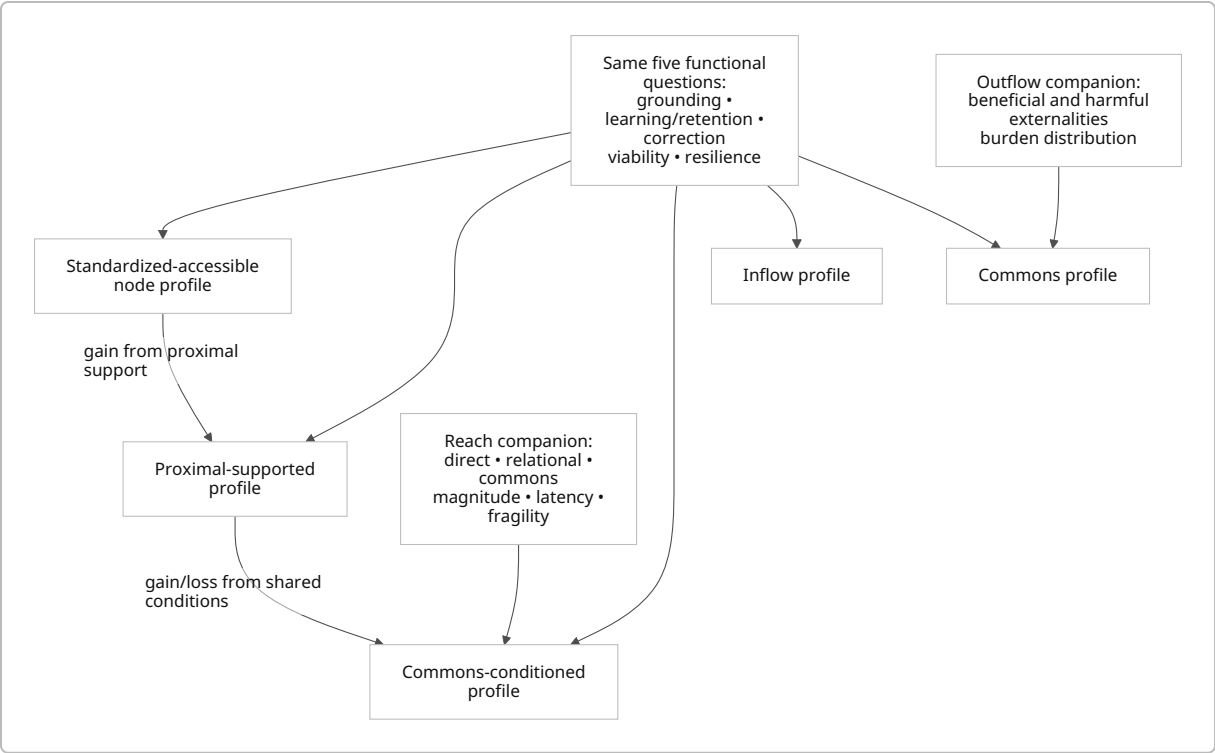
then finite initial reserve predicts increasing risk of crossing the caregiver’s viability boundary unless support, maintenance demand, or care burden changes. This is the formal version of “support that destroys its supporter is not sustainable support.” It should not be equated mechanically with clinical burnout: psychological burnout is multidimensional and would require validated health outcomes, not merely a negative scalar budget.

Synthesis. EbE’s example nevertheless captures a useful measurement warning: **aggregate slack does not identify distributive viability**. National income can rise while a care workforce is exhausted; an AI service can become cheaper while annotators or grid infrastructure bear rising costs; a household’s total resources can be sufficient while one member absorbs nearly all unpaid care. The empirical instrument should therefore retain the distribution of burden, not only its sum.

Candidate measurement design

Synthesis. Measurement should begin with the **accounting boundary**, not the questionnaire. For every assessment state: the focal unit; which relations count as immediate support; which shared systems it depends on; which externalities are included; the time horizon; the tasks or functionings that must be sustained; and which groups have standing to say that a shared condition is failing. Without this boundary, an organization can appear “healthy” merely by moving costs to workers, families, ecosystems, or future periods.

Synthesis. The basic architecture is:



Synthesis. I would avoid the phrase “unsupported individual score.” For disabled people, dementia patients, or people using communication aids, removing essential assistance changes the task and can itself cause harm. The first layer should instead be **standardized-accessible capacity**: performance under a defined, accessible reference environment that retains essential aids. The second is **proximal-supported functioning**, with actual carers, tools, records, retrieval systems, staff, or procedures. The third is **commons-conditioned functioning**, which incorporates current ambient conditions over a specified horizon. The latter two contrasts can often be estimated through natural variation, modeling, or safe cross-over designs rather than deprivation of support. This adapts the ICF’s capacity/performance distinction while avoiding a perverse test of “what happens if we take your wheelchair/carer/memory aid away.” ²³

Component	Person	AI system	Institution
Standardized-accessible node	Ecologically relevant comprehension, learning, calibrated confidence, correction and recovery with necessary aids retained	Frozen model on controlled tasks, known data boundary, fixed compute/interface	Core staff/processes under standardized case mix and defined resources
Proximal-supported	Actual carers, records, assistive tech, clinicians, trusted peers	Retrieval, tools, external memory, human review, monitoring, orchestration	Actual staffing, audit, records, professional networks, escalation routes

Component	Person	AI system	Institution
Commons-conditioned	Functioning under current heat, housing, safety, information environment, care system, public services	Performance under current web/data ecosystem, compute/energy reliability, supply chain, human verification	Performance under current legal, fiscal, information, labour, infrastructure and political conditions
Inflow	Accuracy, timeliness, diversity and reliability of advice; care availability; material support	Data provenance, source diversity, feedback quality, labels, tool reliability, human verification	Evidence quality, frontline reports, complaints, audit information, funding/staff flow
Reach	Direct control; effects on close others; collective civic/environmental effects	Effects on users; downstream models/data; web content; resource demand	Effects on clients, staff, contractors, population, other institutions and future capacity
Commons	Local/national information, care, institutional and ecological conditions on which person depends	Training-data/web commons, verification labour, energy/compute infrastructure, standards and institutions	Information ecosystem, public infrastructure, workforce, fiscal/ecological base, constitutional and professional checks

Synthesis. Each of the five domains can use a common five-level *meaning* while the indicators remain substrate-specific: **0 critical failure, 1 severe/unstable, 2 fragile or only partly adequate, 3 adequate under expected conditions, 4 robust under plausible change**. These are ordinal flags, not numbers to multiply. Raw continuous measures should always be retained.

Synthesis. Candidate indicators are as follows. **Grounding:** for people, accuracy–confidence calibration on relevant decisions and ability to discriminate trustworthy from misleading information; for AI, proper scoring rules, abstention, factuality, calibration and out-of-distribution performance; for institutions, error rates in administrative data, transparent evidence use, reproducibility, independent audit discrepancies, and correction of false public records. AI calibration is already an active technical field: recent work has shown that model training can improve whether human listeners recognize when an answer is likely wrong.

24

Synthesis. Learning and retention: repeated-measures learning slopes plus later retention in humans; learning after controlled updates together with catastrophic-forgetting probes for AI; and whether institutions implement, retain, and later retrieve lessons from incidents, appeals, audits, or policy evaluations. The measure must reward *retained correction*, not rapid volatility.

Synthesis. Correction and independence: mix valid and deliberately invalid challenges. Measure acceptance of valid correction, rejection of misleading correction, time to repair, and diversity of evidence routes. For people this could be adapted experimental feedback tasks. For AI it includes sycophancy tests

and independent retrieval sources. Recent formal and empirical AI work continues to find that preference-based post-training can amplify agreement with user beliefs at the expense of accuracy, making “corrigible but not obedient” a real measurable distinction. ²⁵ For institutions, indicators include appeal success, independent audit, whistleblower protection, response times, judicial review, media plurality, and whether complaints change records or procedures.

Synthesis. Resource viability: person's reliable access to care, housing, sleep, income, medication, communication, and cognitive aids as relevant to the agreed functioning target; AI's compute availability, energy intensity, provenance-preserving data access, human-review capacity and operational budget; institution's staffing, turnover, fiscal reserve, backlog, care capacity, infrastructure and maintenance debt. Resource use must be related to a defined service level. Absolute low resource use is not necessarily health.

Synthesis. Resilience/recovery: expose the system, where ethical, to a controlled shift or use observed shocks. Measure performance loss, recovery time, retained capabilities, and cost of recovery. At commons level this becomes outage recovery, institutional continuity, restoration of trustworthy information, care surge capacity, ecological recovery, or maintenance of service through crisis.

Established. AI makes the commons dependence unusually visible. Shumailov and colleagues demonstrated degradation under repeated replacement of real training data with recursively generated data. Later work showed that collapse is **not inevitable whenever synthetic data are present**: retaining or accumulating real data, changing the data workflow, or adding external correction can stabilize self-consuming training. Provenance and mixture therefore matter more than the slogan “synthetic data cause collapse.” ²⁶ Dataset documentation proposals such as *Datasheets for Datasets* and NIST's AI RMF already treat provenance, documentation, evaluation, and monitoring as important parts of trustworthy AI practice.

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Established. The AI commons is also material and human. The IEA estimates that data centres used about 415 TWh of electricity in 2024, around 1.5% of world electricity use, and projects roughly 945 TWh by 2030 in its base case, with substantial uncertainty. Digital AI and data systems also depend on human platform and microtask labour; ILO surveys document low or precarious earnings and weak social protection for many online platform workers. These inputs belong in a commons-conditioned assessment rather than being hidden outside “model performance.” ²⁸

Synthesis. The **inflow profile** should separately record quantity, accuracy/quality, diversity, independence, latency, continuity, and fragility. A person with ten sources that all copy the same misinformation does not have ten independent routes. Neither does an institution whose ten analysts all rely on one model.

Synthesis. The **reach profile** should report target level, direction, latency, and causal confidence. For example: “high direct reach over medication-taking; moderate relational reach over household routines; extremely low attributable individual reach over national climate, but non-zero collective reach through voting, organization, profession, or consumption.” Reach is a description of causal leverage, **not moral responsibility**.

Synthesis. The **commons profile** should present both central tendency and distribution. At minimum report median, worst-off tail, geographic/group inequality, and concentration of burdens. EbE's aggregate-transfer warning points in the right direction here, but distributional reporting is justified independently by well-established equity and public-health considerations.

Equity, rights, and misuse

Established. Disability frameworks reject the idea that functioning is located only in an impaired body: WHO explicitly describes disability as interaction between health conditions and personal and environmental factors. Capability theory likewise distinguishes having resources from genuinely being able to convert them into valuable functionings. ²⁹ A relational instrument that instead labels assisted functioning as “less authentic” would therefore move backward.

Synthesis. The most important ethical rule is: **entitlement to care must be independent of outflow.** Contribution cannot be a condition for human worth, treatment priority, social protection, or care. A newborn infant, a person with advanced dementia, somebody in coma, or somebody unable to perform paid work does not acquire less claim to support because their measured “contribution” is low. The outflow account is for understanding network sustainability and hidden burdens, not deciding who deserves to live well.

Synthesis. Likewise, caregiver health cannot be hidden inside the recipient's good outcome. A high supported-functioning score for the care recipient paired with a collapsing caregiver profile is a warning, not success. The policy response is not “the carer should contribute less” in a moral sense; it is that the support system has externalized maintenance costs onto one node. Formal care, respite, income protection, accessibility, and shared provision are possible system responses.

Established. Relational-autonomy and care scholarship is particularly relevant because autonomy is not simply independence. The relational-autonomy literature examines how social relationships, oppression, and institutional arrangements can enable or undermine a person's practical autonomy. ³⁰ This is compatible with measuring support without treating dependence itself as a defect.

Synthesis. Do not rank families, neighbourhoods, ethnic groups, or countries by a single “commons-health score.” Such rankings invite stigma and conceal boundary choices. A community facing polluted air, austerity, weak infrastructure, or political exclusion could be misclassified as intrinsically unhealthy when the causal burden lies largely outside its control. Profile the relevant conditions and inequalities instead.

Synthesis. “Healthy information ecosystem” is also politically contestable. Governments could misuse a correction metric to relabel dissent as misinformation. The operational test must therefore concern **process properties**—independent evidence routes, transparent records, appeal, reproducibility, correction of demonstrated error; source plurality—not conformity to an official view. V-Dem's disaggregated approach is preferable to treating democracy as one unchallengeable authority score. ³¹

Established. Trust needs the same caution. OECD data show that perceived voice, responsiveness, evidence use, and institutional trust are related, but high trust is not equivalent to institutional truth or justice. ³²

Synthesis. A healthy epistemic commons may sometimes produce *lower* trust if previously hidden misconduct becomes visible. Measure warranted trust and correctability where possible, not maximum trust.

Synthesis. Goodhart effects are likely. Once appeal resolution time is rewarded, appeals may be closed prematurely. Once factuality scores matter, systems may avoid difficult questions. Once staff productivity is rewarded, unpaid work or quality may disappear from the record. The response should be rotating

indicators, external audits, distributional checks, qualitative challenge routes, and repeated validation against the ultimate outcome rather than permanent optimization of a proxy.

Synthesis. Finally, functional language about AI must not smuggle in claims of moral status. Saying that a model has poor “health” because it is miscalibrated or data-starved is an engineering metaphor. This report found evidence about model reliability and system sustainability, not evidence that current AI systems suffer, have welfare, or possess claims comparable to persons.

What the evidence teaches Evolution by Emergence

Established about the theory. `CORE.md` presents EbE as a model of learning read as a network and explicitly maps learning, maintenance, feedback, transfers, and commons questions across domains. The repository itself also warns that cross-domain correspondences must be demonstrated rather than assumed. `PREDICTIONS.md` now classifies the health-of-intelligence and health-of-commons material as proposed tests that are less developed than the earlier formal results and rest mostly on concepts rather than proofs. That epistemic status is appropriate.

Synthesis — strengthened. Three parts of the meta-model receive substantial independent support. First, **realized functioning is relational**. ICF, capabilities, social determinants, risk sharing, family spillovers, and relational wellbeing all reject a purely isolated-node account. Second, **support can alter functioning without altering the underlying impairment or base model**. Third, **aggregate outcomes can hide who carries the cost**; care economics, distributive health research, and network risk-sharing all make that empirically plausible. ³³

Synthesis — strengthened but narrowed. The idea that persistence requires something like grounding, learning, correction, resource sufficiency, and recovery is also well aligned with several independent literatures. But what the literature supports is **functional analogy**, not a demonstrated universal latent variable called intelligence health. A human mind, an LLM service, a ministry, and a democratic information ecosystem may answer the same higher-order questions while having different mechanisms.

Synthesis — challenged. The **compression** claim needs weakening. Efficient representation and affordable upkeep can be advantageous, but resilience literature repeatedly values diversity, redundancy, slack, modularity, and multiple knowledge systems. The optimal amount of compression is therefore conditional. Excess compression may destroy the very independent correction routes EbE values. ¹⁷

Synthesis — challenged. The proposed **commons** category is too heterogeneous for one ledger unless the ledger is vector-valued. A fishery is subtractable. Knowledge is largely non-rival once produced, but verification and curation are costly. Trust is neither a conserved material nor always good. Care capacity is produced by people who themselves need maintenance. Climate is a coupled physical state, not an institutional fund. The framework needs a resource taxonomy before it can safely transfer equations between cases. Ostrom's work is especially useful here because it insists on analyzing the attributes of the resource and community rather than assuming every commons has the same dilemma. ³⁴

Synthesis — missing. Power needs to become an explicit state variable. “Inflow” says what reaches a node but not who can compel whom, set the accounting boundary, define valid evidence, refuse care, shift costs,

or alter the rules. Political efficacy, relational autonomy, Ostrom's governance principles, and democracy measurement all point toward this omission. 35

Synthesis — missing. Distribution and rights also need first-class status. A viable whole whose surplus depends on one permanently depleted subgroup should not automatically count as a healthy commons. The Lean transfer theorem correctly distinguishes whole and part viability, but a meta-model needs to go further: identify which parts have claims that cannot legitimately be traded against aggregate gain.

Synthesis — missing. Time scale should be explicit. Direct agency can act in seconds; caregiving effects accumulate over months; institutions learn over years; climate and knowledge ecosystems span generations. “Staying in step” needs a horizon T . A system can look healthy at $T = 1$ and doomed at $T = 100$.

Synthesis. Taken together, this suggests a refinement of the central EbE picture:

Realized functioning $_{i,t} = F\Big(\text{capacity}_{i,t}, \text{relational inflows}_{i,t}, \text{shared state}_t, \text{power/access}_{i,t}, \text{history}_{0:t}\Big)$

while

$$\text{shared state}_{t+1} = G(\text{shared state}_t, \{\text{outflows}_{i,t}\}, \text{external shocks}, \text{governance}) .$$

These equations are **Synthesis**, not established mathematical laws. Their main advantage over a scalar product is that they leave interaction structure to empirical testing.

Falsifiable predictions, search record, and annotated bibliography

Hypothesis. The framework becomes scientifically useful only if it predicts something beyond existing capacity measures. The following tests would do that.

Prediction	Design and outcome	What would count against it
Support decomposition predicts realized functioning	Longitudinal studies in people, AI services, and institutions. Estimate standardized-accessible capacity, proximal support, and commons conditions separately. Outcome: later task/service functioning.	Commons and relational variables add no reproducible out-of-sample prediction after node capacity and ordinary covariates.
Independent correction routes predict recovery	Randomize or exploit natural variation in feedback source diversity while holding information amount roughly constant. Outcome: recovery from planted blind spots or real shifts.	Copies of one source recover just as well as genuinely independent routes, with useful effects excluded by confidence intervals.

Prediction	Design and outcome	What would count against it
Care that exceeds regeneration produces delayed dual decline	Prospective caregiver–recipient cohorts measuring recipient function, caregiver reserves/health, formal support and time use.	Sustained high care loads do not predict caregiver deterioration or later recipient instability after confounding and selection are addressed.
Commons condition moderates node capacity	Multi-site/person or multi-deployment AI studies with the same baseline capacity exposed to different heat, misinformation, care, provenance, compute or institutional environments. Test capacity × environment interactions.	Effects are fully additive or negligible; no stable moderation appears across preregistered replications.
Real-data/provenance routes prevent recursive AI degradation	Controlled successive training under replacement, accumulation and provenance-filtered data regimes. Outcome: tail coverage, calibration and held-out real-distribution performance.	Provenance-preserving and independently grounded regimes deteriorate at the same rate as pure recursive replacement.
Distributional viability predicts failures missed by averages	Track organizations, care systems or communities with similar aggregate resources but different concentration of workload/slack. Outcome: local service failure, staff exit or dependency collapse.	Aggregate slack predicts failure as well as or better than distributional minima/quantiles across external samples.

Synthesis. A particularly decisive test of the “one profile at three levels” claim would ask whether the five domain scores show **predictive homology**, not numerical identity: for example, whether correction-route integrity independently predicts post-shift recovery in humans, AI systems, and institutions after controlling for baseline competence. Failure in one substrate would not destroy the whole framework, but it would argue against substrate neutrality.

Established search record. I conducted a structured, non-systematic review through 7 October 2026. I retrieved records primarily from publisher pages, PubMed, WHO, OECD, UNEP, HM Treasury, NIST, IEA, ILO, V-Dem, PMLR/NeurIPS/ACL, and the EbE GitHub repository. Search families included combinations of *capability conversion factors personal social environmental*; *ICF capacity performance facilitators barriers*; *family spillover caregiver QALY*; *relational wellbeing*; *collective efficacy/social capital health*; *current wellbeing capital stocks future wellbeing*; *inclusive wealth/Dasgupta*; *knowledge commons/Ostrom*; *epistemic health*; *political efficacy/V-Dem/trust*; *risk-sharing social networks*; *network contagion homophily*; *heat cognition, allostatic load cognition, loneliness cognitive decline, misinformation exposure*; *model collapse recursive synthetic data, sycophancy/RLHF, calibration, dataset provenance, AI energy, digital platform workers*; *viability theory*; and the exact Lean theorem names supplied in the brief. Bibliographic details below were checked against the retrieved record, not recalled from memory.

Established search limits. This was not a PRISMA systematic review. I did not run exhaustive subscription-database searches of Scopus, Web of Science, Embase, PsycINFO, or EconLit, and I did not perform exhaustive backward/forward citation screening. English sources were prioritized. Several papers were available only as publisher or PubMed abstracts; those are marked below. One QALY paper was available in

full as an uploaded PDF. The brief refers to two previous reports, but both were not available to me as full documents during this run, so I treated the brief's summary of them as background rather than evidence. Consequently, “not found” must not be read as proof of novelty.

Annotated bibliography

Source	Why it matters and access verified
Evolution by Emergence, <code>CORE.md</code> .	Theory under assessment, not evidence. Defines the network/learning/ledger model and its proposed cross-domain interpretation. Repository file retrieved directly.
Evolution by Emergence, <code>PREDICTIONS.md</code> .	Theory under assessment. Explicitly distinguishes empirical predictions from Lean results and describes health-of-intelligence/commons proposals as comparatively undeveloped. Retrieved directly.
Evolution by Emergence, <code>NetworkVortexLedger.lean</code> .	Contains and proves the two transfer theorems discussed here; crucially states their static, linear, additive scope. Source retrieved directly; I did not independently rebuild the Lean project.
WHO. International Classification of Functioning, Disability and Health.	Closest operational precedent for separating functioning from environmental facilitators/barriers and for measuring at person and population levels. Official WHO source. ³⁶
WHO. <i>How to Use the ICF: A Practical Manual</i> (2013).	Practical implementation source for contextual functioning measurement. Official record retrieved. ³⁷
Robeyns I. “The Capability Approach: a theoretical survey.” 2005. doi: 10.1080/146498805200034266	Core account of capabilities versus functionings and multidimensional wellbeing. Publisher full/record retrieved. ³⁸
Robeyns I. “The Capability Approach in Practice.” 2006. doi:10.1111/j.1467-9760.2006.00263.x	Useful for operational issues and heterogeneity in turning resources into valued functionings. Publisher record/full-text page retrieved. ³⁹
Coast J et al. “Valuing the ICECAP capability index for older people.” 2008. doi:10.1016/j.socscimed.2008.05.015	Shows a health-economic instrument grounded in broader capability rather than health alone. PubMed abstract/record checked. ⁴⁰
White SC. “Analysing wellbeing: a framework for development practice.” 2010. doi: 10.1080/09614520903564199	Probably the closest broad relational ontology: material, relational, subjective; individual and collective. Publisher abstract/record checked. ⁴¹

Source	Why it matters and access verified
White SC, Jha S. "Exploring the Relational in Relational Wellbeing." 2023. doi:10.3390/socsci12110600	Develops persons, context, power and relationships as mutually constitutive, useful for correcting a node-centric model. Full article/record retrieved. ⁴²
WHO Commission on Social Determinants of Health. <i>Closing the Gap in a Generation</i>. 2008. ISBN 9789241563703	Establishes the broad public-health tradition of locating health in living and structural conditions rather than only in individuals. Official WHO record. ¹⁰
Spencer A et al. "The QALY at 50: One story many voices." 2022. doi:10.1016/j.socscimed.2021.114653	Historical review documenting QALY methodological debates, spillovers, capability alternatives and equity issues. Full text supplied by user.  turn0file0 
Al-Janabi H et al. "A framework for including family health spillovers in economic evaluation." 2016. doi: 10.1177/0272989X15605094	Direct precedent for measuring effects beyond the focal patient. PubMed abstract/record checked. ⁴³
Zarit Burden Interview validation literature.	Demonstrates that the health cost borne by carers is measurable and cannot safely be treated as free support. PubMed abstracts checked. ⁶
Mackenzie C, Stoljar N, eds. <i>Relational Autonomy</i>. 2000. doi:10.1093/oso/9780195123333.001.0001	Important corrective to equating autonomy with independence; social relations may enable or impair agency. Oxford publisher record/preview checked. ³⁰
Sampson RJ, Raudenbush SW, Earls F. "Neighborhoods and violent crime: a multilevel study of collective efficacy." 1997. doi:10.1126/science.277.5328.918	Canonical operationalization of collective efficacy as a community-level relational property. Publisher/PubMed-linked record checked. ⁴⁴
Villalonga-Olives E, Kawachi I. Review of adverse effects of social capital. 2017. doi: 10.1016/j.socscimed.2017.10.020	Critical because connection and trust can transmit exclusion and harm; "more social capital" is not automatically better. Publisher abstract checked. ⁴⁵
OECD. <i>How's Life?</i> wellbeing framework.	Separates current wellbeing from natural, human, economic and social resources for future wellbeing. Official OECD source. ⁴⁶
UNEP. <i>Inclusive Wealth Report 2023</i>.	Makes sustaining productive capital stocks explicit in national wellbeing accounting. Official UNEP source. ⁴⁷

Source	Why it matters and access verified
Dasgupta P. <i>The Economics of Biodiversity: The Dasgupta Review</i>. 2021.	Strong stock/flow analogy for nature, while keeping ecological assets distinct from current income. Official UK government record. ⁴⁸
Whitmee S et al. "Safeguarding human health in the Anthropocene epoch." 2015. doi:10.1016/S0140-6736(15)60901-1	Foundational planetary-health report linking human health to Earth-system conditions. PubMed/publisher abstract checked. ⁴⁹
Ostrom E. <i>Governing the Commons</i>. 1990.	Core evidence and theory on long-lived governance of common-pool resources; emphasizes monitoring, participation, conflict resolution and fit to context. Cambridge publisher record checked. ⁵⁰
Hess C, Ostrom E, eds. <i>Understanding Knowledge as a Commons</i>. 2007/2011 edition.	Essential for distinguishing knowledge from subtractable natural-resource commons and focusing on access, preservation, verification and institutions. MIT Press record. ⁵¹
Piovarchy A, Siskind A. "Epistemic Health, Epistemic Immunity and Epistemic Inoculation." 2023. doi:10.1007/s11098-023-01993-9	Closest conceptual prior art to "health" across persons, communities and nations, but not a cross-substrate instrument. PubMed/publisher abstract checked. ⁹
V-Dem Dataset v16. 2026.	Provides disaggregated indicators for democratic institutions, participation, deliberation and constraints; possible commons-correction data source. Official dataset record. ⁵²
OECD. <i>Survey on Drivers of Trust in Public Institutions</i>, 2024/2026 data.	Measures voice, responsiveness, reliability, evidence use and trust; useful for inflow/reach but not itself a truth measure. Official OECD data/report. ⁵³
Christakis NA, Fowler JH. "The Spread of Obesity in a Large Social Network over 32 Years." 2007. doi:10.1056/NEJMsa066082	Canonical network-health study; valuable mainly alongside its causal-identification limitations. Full NEJM page retrieved. ⁵⁴
"The Spread of Obesity in a Social Network" correspondence. 2007. doi:10.1056/NEJMc072478	Contemporary critique raising social homogeneity and socioeconomic confounding, illustrating why network association is not causal reach. NEJM page retrieved. ⁵⁵
Ambrus A, Mobius M, Szeidl A. "Consumption Risk-Sharing in Social Networks." 2014. doi:10.1257/aer.104.1.149	Formal and empirical bridge from network architecture to informal insurance; more realistic than unrestricted pure transfers. AEA full record. ⁵⁶

Source	Why it matters and access verified
Townsend RM. "Consumption Insurance." 1995. doi:10.1257/jep.9.3.83	Establishes empirical benchmarks for partial versus complete informal insurance. AEA full record. ⁵⁷
Alatas V et al. "Network Structure and the Aggregation of Information." 2016. doi:10.1257/aer.20140705	Shows how network structure affects information aggregation in real communities. AEA record with replication material. ⁵⁸
Folke C et al. "Adaptive Governance of Social-Ecological Systems." 2005. doi:10.1146/annurev.energy.30.050504.144511	Connects resilience to learning, diverse knowledge, networks and bridging organizations; important counterweight to maximizing compression. Publisher abstract/full record. ¹⁷
Aubin J-P, Bayen AM, Saint-Pierre P. Viability Theory: New Directions. 2011. doi: 10.1007/978-3-642-16684-6	Formal framework for keeping trajectories within constraints over time; strongest mathematical analogue for dynamic commons viability. Springer record/preview. ⁵⁹
Taylor L et al./review, "The Impact of Environmental Stress on Cognitive Performance." 2019.	Review of heat, cold, altitude and other environmental stressors; supports environment-cognition effects while documenting heterogeneity. PubMed abstract checked. ⁶⁰
"The association between allostatic load and cognitive function: a systematic and meta-analytic review." 2020.	Finds small associations of physiological stress burden with global cognition and executive function; important evidence but not strong enough for a universal multiplier. PubMed abstract checked. ⁶¹
"Association between loneliness and mild cognitive impairment ... meta-analysis." 2024.	Supports a longitudinal association while illustrating heterogeneity and residual uncertainty around relational exposures and cognition. PubMed abstract checked. ⁶²
Meta-analysis of health-misinformation exposure. 2024. doi:10.1177/09636625241266150	Randomized-trial synthesis finds a small average increase in misbelief after exposure, directly relevant to informational inflow. PubMed abstract checked. ⁶³
NIST. Artificial Intelligence Risk Management Framework 1.0. 2023. doi:10.6028/NIST.AI.100-1	Cross-sector framework covering governance, measurement, management and risks to individuals, organizations and society; not a model-welfare instrument. Official NIST source. ⁶⁴
Gebru T et al. "Datasheets for Datasets." 2021. doi:10.1145/3458723	Formalizes dataset provenance, composition, collection and intended-use documentation. Published CACM status verified; arXiv full text available. ⁶⁵

Source	Why it matters and access verified
Shumailov I et al. “AI models collapse when trained on recursively generated data.” 2024. doi:10.1038/s41586-024-07566-y	Demonstrates degradation in specified recursive replacement regimes. Full Nature article page and later correction verified. ⁶⁶
Kazdan J et al. “Collapse or Thrive: Perils and Promises of Synthetic Data in a Self-Generating World.” ICML 2025.	Shows model-collapse outcomes depend on training workflow; accumulation of real data can avoid divergent collapse. Published PMLR paper, not merely the earlier preprint. ⁶⁷
Gillman N et al. “Self-Correcting Self-Consuming Loops for Generative Model Training.” ICML 2024.	Shows external/knowledge-based correction can stabilize self-consuming loops in studied settings. Published PMLR paper. ⁶⁸
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“LACIE: Listener-Aware Finetuning for Calibration in Large Language Models.” NeurIPS 2024. doi:10.52202/079017-1364	Demonstrates calibration as communication to downstream humans, fitting the relational rather than purely internal view of model quality. Published conference record. ²⁴
IEA. <i>Energy and AI</i>. 2025.	Best official source retrieved on the material energy dependence of AI/data-centre infrastructure and uncertainty in future demand. Official report/data source. ⁷⁰
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Synthesis. The literature therefore supports a final reframing: the promising object is not a universal “EQ-5D for intelligence” in which humans, models, governments, and ecosystems receive comparable total scores. It is a **shared interface for asking the same causal questions at different levels**: *What can this node do? What makes that functioning possible? What information and support reach it? What shared conditions does it depend on? Can error enter and be repaired? What does its activity do to others and to those shared conditions? How far does that effect actually reach? Who carries the maintenance cost? And does the arrangement still work after the world changes?* Existing disciplines answer pieces of those questions. The potentially new research programme is to make the pieces interoperable, then test whether that interoperability predicts persistence and recovery better than measuring the node alone.

¹ ² ³³ ³⁶ International Classification of Functioning, Disability and Health (ICF)

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